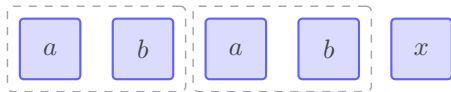


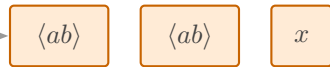
example before: $N_t = 5$ positions



two non-overlapping (a, b) matches: $r_t = 2$

BPE merge
 $(a, b) \mapsto \langle ab \rangle$

example after: $N_{t+1} = 3$ positions



each framed pair becomes one $\langle ab \rangle$ token

one merge rule spends one vocabulary entry to buy r_t fewer weighted positions

spend: one fresh vocabulary entry

$$V_{t+1} = V_t + 1$$

buy: r_t fewer weighted positions

$$N_{t+1} = N_t - r_t$$